

ANWAR UDIN SAYFULLOH

AI Engineer | Machine Learning | Data

Engineering Graduate (Geophysics) - Institut Teknologi Sumatera, Class of 2026

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PROFESSIONAL SUMMARY

Engineering graduate (B.Eng. Geophysics, Institut Teknologi Sumatera, GPA 3.31/4.00) with hands-on experience across Python-based data analysis, machine learning, deep learning, and scientific computing. Proven technical leadership as Team Lead for a data-driven mineral exploration proposal developed for PT Antam Tbk, coordinating a four-person team from technical analysis through final deliverable. Skilled at converting ambiguous, real-world problems into structured, data-driven solutions across geoscience, computer vision, and predictive analytics. Combines strong quantitative reasoning, cross-functional collaboration, and fast adaptability - well suited for Data/AI Engineering.

KEY STRENGTHS

Data-Driven Problem Solving | Machine Learning & Deep Learning | Team Leadership & Coordination | Cross-Functional Collaboration | Analytical & Quantitative Reasoning | Adaptability & Fast Learning | Project Ownership & Reporting

TECHNICAL SKILLS

Programming Languages: Python, SQL, C++, HTML, CSS, JavaScript

Data Analysis & Machine Learning: Pandas, NumPy, SciPy, Scikit-learn, XGBoost, Random Forest, K-Means, KNN, DBSCAN, PCA, SMOTE, GridSearchCV, Feature Engineering, EDA, Statistical Analysis

Deep Learning & Computer Vision: PyTorch, TensorFlow/Keras, CNN, ResNet34, EfficientNet, YOLOv8, YOLO11, OpenCV, Transfer Learning, Data Augmentation

Tools & Platforms: Git, GitHub, Streamlit, Jupyter Notebook, Numba, Joblib, QGIS

Currently Developing (AI Engineering Program): RAG Architecture, Agentic AI, LLM Integration, REST API, FastAPI, Docker, MLflow

Geoscience Software: SonarWiz, Golden Software Surfer, Oasis Montaj, Petrel, MATLAB

PROFESSIONAL EXPERIENCE

PT Qyudos Geosurvey Indonesia - Marine Geophysics Intern

Bandung, Indonesia | Jul 2025 - Aug 2025

- Processed Sub-Bottom Profiler (SBP) data for subsea pipeline route analysis, identifying sedimentary layers and potential buried hazards.
- Conducted Side Scan Sonar (SSS) acquisition for seafloor imaging and interpretation to support pipeline routing decisions.
- Performed real-time interpretation during marine geophysical acquisition and supported equipment calibration and acquisition quality control.
- Processed and interpreted survey data using SonarWiz and Golden Software Surfer.

Sahara AI - AI Data Annotator (Remote)

Remote | Dec 2024 - May 2025

- Annotated image, text, and video datasets according to project-specific labeling guidelines for AI model training.
- Reviewed annotated data and verified labeling consistency, identifying and correcting quality inconsistencies.
- Produced instructional videos documenting annotation workflows and software procedures for team reference.

LEADERSHIP EXPERIENCE

GARAGA Team, ANTAM Hackathon (Unit Geomin) - Team Lead / Technical Lead

2026

- Led a 4-person technical team in developing a data-driven mineral prospectivity mapping proposal for PT Antam Tbk, integrating geological, geochemical, geophysical, and spatial evidence with machine learning (XGBoost).
- Directed technical strategy, including evaluation of geostatistical uncertainty-quantification methods from published research, and resolved a critical data gap by designing a preprocessing workaround.

- Coordinated task allocation, cross-functional technical discussion, and final deliverable integration, ensuring all submission requirements and deadlines were met.

SELECTED AI / DATA / ML PROJECTS

1D Magnetotelluric Inversion & Uncertainty Analysis - Python, Numba, Joblib

Research Publication (ResearchGate) | 2026

- Built an end-to-end Python workflow for 1D MT forward modeling, inversion, and uncertainty analysis using Sambridge's Neighborhood Algorithm, benchmarked against Levenberg-Marquardt.
- Achieved up to 55.8x computational speedup via Numba JIT compilation across one million 3-layer forward model evaluations.
- Applied the workflow to field data from Kilauea Volcano, achieving a ~21% lower inversion misfit (1.99 vs. 2.52) than Levenberg-Marquardt while quantifying model uncertainty through posterior probability analysis.

Gold Deposit Mapping / Mineral Prospectivity - Python, XGBoost, QGIS

2026

- Developed a mineral prospectivity mapping workflow integrating geological, geochemical (39 evidence layers), geophysical, and radiometric data using XGBoost trained on approximately 8 million rows.
- Generated exploration-target probability maps with spatial uncertainty to support geological decision-making.

Rock Classification Using Deep Learning - PyTorch, ResNet34, YOLO, Streamlit

2026

- Trained a 53-class rock image classifier (2,343 images) using a custom CNN and transfer learning with ResNet34, deployed via an interactive Streamlit application.
- Fine-tuned a YOLO11 detection model for automated rock-sample cropping, reaching approximately 0.99 mAP50 and 0.85 mAP50-95.

Landslide Risk Prediction - Python, XGBoost, Random Forest, SMOTE

2025 - 2026

- Built a landslide risk classification model using StandardScaler, SMOTE, and GridSearchCV-tuned XGBoost and Random Forest, evaluated via accuracy, precision, recall, F1-score, and confusion matrix.

Subsurface Utility Detection on GPR - YOLO, OpenCV, Roboflow

2025

- Developed an automated computer vision pipeline detecting GPR hyperbola reflections from underground pipes and utilities using a custom-trained YOLO model with tailored data augmentation.

EDUCATION

Institut Teknologi Sumatera (ITERA) - B.Eng. (S.T.), Geophysics

Lampung, Indonesia | 2022 - 2026 | GPA: 3.31 / 4.00

- Thesis: "Uncertainty Analysis of 1D Magnetotelluric Inversion Model Using the Neighborhood Algorithm: Application to Kilauea Volcano"
- Relevant coursework: Exploration Geology, Geophysical Methods, Geophysical Data Processing, Geophysical Modeling, Remote Sensing

Purwadhika Digital Technology School - AI Engineering Program

Indonesia | Aug 2026 - Present

- Focus: Python for AI, data processing, machine learning, SQL, AI backend & REST API, RAG architecture, agentic AI, software engineering best practices

CERTIFICATIONS

- DataCamp - Python Developer Associate (September 2026)
- DataCamp - SQL Associate (September 2026)
- IBM SkillsBuild / Hacktiv8 Indonesia - Data Classification and Summarization, Student Developer Initiative (2025)
- ITERA Language Center - English Proficiency Test (TOEFL), Score 590 (June 2026)

LANGUAGES

Indonesian - Native | English - Working Proficiency (TOEFL 590)